Week 8: Endowments, winners, and support for protection

ECON 342 International Trade

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Week 8: Who gains and who loses from trade?

THE QUESTION

Trade can raise national income while lowering the income of particular workers, owners, sectors, and regions. One model family—Heckscher–Ohlin and Stolper–Samuelson—organizes those distributional effects.

BY THE END OF CLASS

- distinguish factor abundance from factor intensity
- state the Heckscher–Ohlin prediction
- calculate a Stolper–Samuelson price effect
- separate the H–O trade prediction from the Stolper–Samuelson distribution prediction
- explain distributional conflict from trade and AI

CLASS MAP

10:30	Canadian endowments
10:50	H–O and factor prices
11:20	H–O policy screen
11:35	ten-minute break
11:45	Canadian evidence and AI
12:05	political economy

Aggregate gains and individual losses can coexist

NATIONAL ACCOUNTING

Lower prices, export access, productivity, and variety can raise total real income.

The gains-from-trade result

A positive aggregate surplus means the winners could in principle compensate the losers.

HOUSEHOLDS AND PLACES

Compensation may be incomplete. A displaced worker, a specialized plant, or a town built around one industry can bear a large transition cost.

The distribution question

Who receives the gains, who bears the loss, and how quickly can labour and capital move?

YOUR MOVE Why does "Canada gains" fail to answer whether every Canadian gains?

Why Heckscher–Ohlin matters

RICARDO: GAINS FROM RELATIVE COSTS

Comparative advantage says that a country tends to export goods with a lower opportunity cost. Specialization and exchange can then raise total real income.

H–O: ENDOWMENTS AND OWNERSHIP

Countries have different relative supplies of productive factors, and goods use those factors in different proportions. This gives comparative advantage an endowment-based mechanism.

What Ricardo leaves open

Why do relative costs differ across countries? Who owns the resources that become more or less valuable after trade?

The distribution extension

People earn wages and rents from factors. Stolper–Samuelson connects a change in goods prices to the real incomes of factor owners.

Why we care: H–O supplies one source of comparative advantage; paired with Stolper–Samuelson, it identifies potential long-run winners and losers.

YOUR MOVE What question about trade can H–O answer that Ricardo alone leaves open?

Canada combines natural-resource abundance with skills

369M ha
of forest, more than 9 hectares per person

32.8%
of world potash production in 2024

64%
of adults aged 25–64 held a college or university credential in 2025

An endowment is a starting point, not a complete explanation

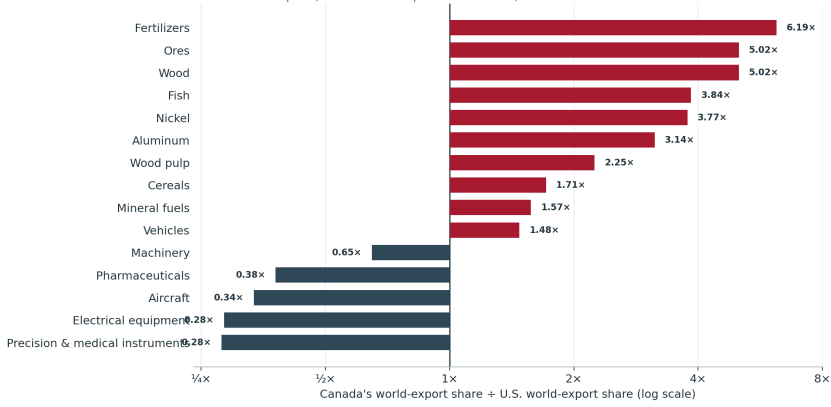
Resources and skills influence relative production costs. Technology, infrastructure, scale, policy, geography, and firm capabilities also shape the observed export basket.

Source: Natural Resources Canada, State of Canada's Forests 2025 and Potash Facts; Statistics Canada Table 37-10-0130-01.

YOUR MOVE Which endowment could support forestry exports? Which could support commercial services or aerospace?

Canada is relatively specialized in resources and land-intensive products

Selected HS chapters, merchandise exports to the world, 2023



Source: UN Comtrade API, HS 2017, preview endpoint. Revealed specialization reflects prices, policy, geography and supply chains.

YOUR MOVE Which bars fit a land, resource, or energy-abundance explanation? Which bars require other mechanisms too?

Heckscher–Ohlin in plain English

WHY ENDOWMENTS CAN SHAPE TRADE

1. Countries differ in their **relative supplies** of land, labour, capital, resources, and skills.
2. A relatively plentiful factor usually has a **lower relative price**.
3. Goods use factors in different proportions. Goods that intensively use the abundant factor tend to have a **lower relative cost**.
4. The country tends to **export those goods** and import goods that intensively use its scarce factors.

CANADA EXAMPLE

Canada has abundant land and natural-resource assets relative to its population. The model therefore predicts relatively strong exports of cereals, lumber, fertilizers, ores, and energy.

The income link

People own factors. Export demand raises the return to the factor used intensively in the export good. Stolper–Samuelson formalizes this long-run income effect.

YOUR MOVE Which factor is abundant? Which good uses it intensively? What trade pattern follows?

Canada and South Korea: a useful H–O contrast

0.92 ha
arable land per person in Canada, 2024

0.025 ha
arable land per person in South Korea, 2024

36.7×
Canada's land-per-person ratio relative to Korea

CANADIAN EXPORTS TO KOREA, 2025

Product section	Share
Energy	32.8%
Ores and non-metallic minerals	21.6%
Farm, fishing, and intermediate food	9.2%
Forestry and building materials	4.8%
Four land/resource groups	**68.4%**

CANADIAN IMPORTS FROM KOREA, 2025

Product section	Share
Motor vehicles and parts	50.9%
Industrial machinery and parts	11.1%
Electronic and electrical equipment	9.8%
Three manufacturing groups	**71.8%**

Interpretation: Canada is relatively land and resource abundant. Korea is land scarce and organized around capital- and technology-intensive manufacturing.

Consistent with H–O, not proof: technology, scale, industrial policy, firms, and supply chains also shape this pattern.

Source: World Bank, arable land per person, 2024; Statistics Canada Table 12-10-0171-01, customs-basis goods trade, 2025. Shares calculated from values in current Canadian dollars.

Factor abundance and factor intensity are different ratios

COUNTRY COMPARISON

Canada is relatively land abundant if

$$\left(\frac{T}{L}\right)_C > \left(\frac{T}{L}\right)_P.$$

This compares factor supplies across countries.

INDUSTRY COMPARISON

Grain is land intensive if

$$\left(\frac{T}{L}\right)_G > \left(\frac{T}{L}\right)_A.$$

This compares production methods across goods at common factor prices.

Abundance describes a country. Intensity describes a good. Both are relative concepts.

YOUR MOVE Why does "Canada has more workers" fail to establish labour abundance?

A two-country, two-good classification

FACTOR ENDOWMENTS

Country	Land (T)	Labour (L)	(T/L)
Canada	120	40	**3.0**
Partner	60	60	1.0

Canada is relatively land abundant.

FACTOR REQUIREMENTS PER UNIT

Good	Land	Labour	(T/L)
Grain	4	2	**2.0**
Apparel	1	2	0.5

Grain is relatively land intensive.

Illustration: these are teaching numbers, not measured Canadian coefficients.

YOUR MOVE State the H–O prediction for the direction of Canada's trade.

H–O is a hypothesis, not a label for every Canadian export

CLOSE FIT

- potash and ores use mineral deposits
- cereals use land and transport infrastructure
- forestry uses forest resources and mills
- some services use a highly educated workforce

NEEDS OTHER MODELS

- two-way auto trade reflects scale and supply chains
- aerospace reflects clusters and accumulated capabilities
- only some firms export within each industry
- distance and market size shape every flow

Observed trade reflects endowments, technology, scale, firms, geography, history, and policy together.

YOUR MOVE Why would a failure of one H–O prediction not imply that endowments never matter?

Trade changes product prices and therefore factor demand

EXPORT OPPORTUNITY

Foreign demand raises the relative price of Canada's export good. Firms producing it demand more of the factors used intensively in that sector.

IMPORT COMPETITION

Access to lower-priced imports reduces the relative price received by domestic producers of the import-competing good.

Product prices $\longrightarrow$ factor demand $\longrightarrow$ wages and rents

Trade affects income distribution even when every household can buy the cheaper imported good.

YOUR MOVE For land-abundant Canada, which factor's return should rise when the relative price of grain rises?

A factor-price calculation before trade

Competitive firms earn zero economic profit. Unit prices equal unit factor costs:

$$\underbrace{p_G}_{6} = 2r + w, \qquad \underbrace{p_A}_{6} = r + 2w.$$

READ THE COEFFICIENTS

One unit of grain uses two units of land services and one unit of labour. Apparel uses one unit of land services and two units of labour.

SOLVE

$$2r + w = 6, \qquad r + 2w = 6$$

r=2, w=2.

Toy model: fixed unit requirements make the factor-price arithmetic transparent.

YOUR MOVE Verify the solution in both cost equations.

A higher grain price magnifies the return to land

Trade raises the grain price from 6 to 8 while the apparel price stays at 6:

$$2r + w = 8, \qquad r + 2w = 6.$$

NEW FACTOR PRICES

$$\mathbf{r = \frac{10}{3} = 3.33}, \qquad \mathbf{w = \frac{4}{3} = 1.33}.$$

PERCENTAGE CHANGES

$$\frac{3.33 - 2}{2} \times 100 = \mathbf{+66.7\%},$$

$$\frac{1.33 - 2}{2} \times 100 = \mathbf{-33.3\%}.$$

Grain rose 33.3%. Land rent rose more; the wage fell in terms of both goods.

YOUR MOVE Why must apparel's unchanged price restrict the combination of r and w?

Stolper–Samuelson turns goods prices into real factor incomes

EXPORT-GOOD PRICE RISES

The real return to the factor used intensively in the export good rises. In the two-good model, it rises relative to both goods.

OTHER FACTOR

The real return to the other factor falls. Cheaper imports do not guarantee that its owners gain overall.

Political implication

Owners of relatively scarce factors can oppose openness even when national real income rises.

Scope: the exact theorem assumes two goods, two factors, common technology, competitive markets, and both goods produced. Treat it as disciplined intuition, not a Canadian point forecast.

YOUR MOVE In the toy model, who gains and who loses from the grain-price increase?

Use one H–O chain to answer four questions

Question	H–O / Stolper–Samuelson logic	Canadian application
Trade pattern	relative abundance × factor intensity	why land- and resource-intensive goods are prominent exports
Goods-price shock	identify the factor used intensively in the good whose price changes	which factor return should rise or fall
Distribution	map factor payments into ownership and household income	who gains or loses in real purchasing power
Evidence check	test assumptions and competing forces	technology, trade costs, input links, ownership, and adjustment frictions

YOUR MOVE Which row explains trade composition? Which row explains the distribution of income?

The H–O mechanism is a chain of relative comparisons

1. **Compare endowments:** which factor is relatively abundant in Canada?
2. **Compare techniques:** which good uses that factor relatively intensively?
3. **Predict trade:** H–O predicts exports of the abundant-factor-intensive good.
4. **Trace income:** if trade raises that good's relative price, Stolper–Samuelson predicts a higher real return to its intensive factor and a lower real return to the other factor.

Relative endowments $\longrightarrow$ relative costs $\longrightarrow$ trade pattern $\longrightarrow$ goods prices $\longrightarrow$ real factor returns

Where evidence enters

The formal model gives a disciplined benchmark. Technology, ownership, trade costs, input links, and slow adjustment determine how closely an observed Canadian case follows it.

YOUR MOVE At which arrow would a new tariff enter the chain?

A tariff changes factor returns through the domestic goods price

Begin from the open-economy prices in the toy model:

$$p_G = 8, \quad p_A = 6 \quad \Longrightarrow \quad r = \frac{10}{3}, \quad w = \frac{4}{3}.$$

A tariff on labour-intensive apparel raises its domestic price to $p_A = 7$, while $p_G = 8$:

SOLVE THE NEW ZERO-PROFIT SYSTEM

$$2r + w = 8, \qquad r + 2w = 7$$

$$\mathbf{r = 3}, \qquad \mathbf{w = 2}.$$

READ THE INCIDENCE

Apparel's price rises 16.7%. The wage rises 50%, while land rent falls 10%. This is the Stolper–Samuelson magnification effect in the opposite direction from greater openness to grain trade.

What this does not establish: whether the tariff raises national welfare. Tariff revenue, consumption distortion, retaliation, input costs, and adjustment remain outside this calculation.

YOUR MOVE Verify $r = 3, w = 2$ in both equations. Who gains in real terms in this toy model?

The short-run winner depends on the shock and the input link

Shock	Likely direct winner	Likely direct loser	Important spillover
better potash export access	mines, deposits, rail-linked suppliers	competing foreign producers	fertilizer buyers may face a higher price
tariff on imported steel	domestic primary steel producers	steel importers	autos and machinery pay more for inputs
foreign duty on lumber	foreign competing mills	Canadian mills and forest towns	builders abroad pay more
tighter dairy import quota	quota holders and protected farms	importers and consumers	food processors face restricted sourcing
stricter auto rules of origin	qualifying regional suppliers	non-qualifying sourcing networks	compliance raises production cost

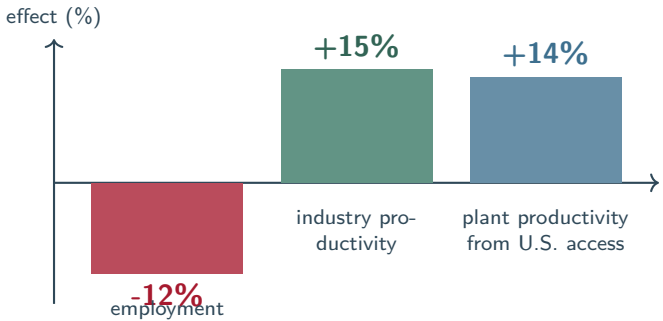
These are first-round predictions. Exchange rates, retaliation, contracts, inventories, and policy responses can change the final incidence.

YOUR MOVE Choose one row, identify the factor used intensively, and name one adjustment friction that requires evidence beyond H–O.

Ten-minute break

At 11:45 we test the H–O benchmark against Canadian adjustment and political support.

The FTA produced contraction and productivity gains together



CANADIAN TARIFF CUTS

In the most exposed industries, low-productivity contraction reduced employment while raising industry productivity.

U.S. TARIFF CUTS

Better foreign access raised Canadian plant productivity in the most exposed industries.

Source: Trefler, American Economic Review (2004). Effects refer to the industries with the deepest respective tariff cuts, not all manufacturing.

YOUR MOVE Why are the employment and productivity estimates not contradictory?

Canadian workers adjusted across sectors over sixteen years



The bilateral-access qualification

Canadian tariff cuts created import competition, while U.S. tariff cuts created alternative export opportunities. The two effects partly offset each other.

"Little average long-run effect" does not mean every worker recovered fully or that every trade shock permits easy mobility.

Source: Kovak and Morrow, Review of Economic Studies 92(6), 2025; matched Canadian worker records, 1984–2004.

YOUR MOVE Which model assumption fails during the initial layoff and transition?

Recent Canadian employment losses are concentrated in exposed industries

-1.4%
payroll employment in industries highly dependent on U.S. demand, Dec. 2024–Aug. 2025

+0.2%
payroll employment in less-dependent industries over the same period

-7.8%
motor-vehicle-parts employment, a loss of 5,600 jobs

Descriptive evidence: exposed employment had already weakened during 2024. These changes should not be presented as a clean causal tariff estimate.

Source: Statistics Canada, Economic and Social Reports, "Recent employment trends in industries dependent on U.S. demand," December 2025.

YOUR MOVE Why does a national employment total hide the adjustment mechanism?

Canada contains several regional trade economies

Region	Export strengths	Endowments and capabilities	Main exposure
Prairies	grain, oilseeds, potash	land, mineral deposits, rail terminals	foreign market access and transport
Alberta	energy and petrochemicals	reserves, pipelines, processing plants	U.S. demand and route concentration
British Columbia	forestry, minerals, energy	forests, mills, ports, local skills	lumber duties and commodity cycles
Ontario	autos, parts, steel, machinery	plants, supplier networks, tooling	tariffs, rules of origin, border delays
Quebec	aluminum, aerospace, services	hydro power, smelters, engineering clusters	tariffs and global investment cycles
Atlantic Canada	fisheries, food, energy	marine resources, ports, processing	market access and standards

A national policy can create different provincial coalitions because endowments, routes, and industrial structures differ.

Source: Illustrative synthesis from Statistics Canada provincial merchandise exports and Natural Resources Canada sector evidence.

YOUR MOVE Choose one region and predict its strongest short-run trade-policy concern.

AI changes which human tasks remain scarce

THE H–O CONNECTION

Relative factor supplies and production techniques shape comparative advantage. AI and robots change the tasks machines can perform, so the demand for different human capabilities can change.

THREE FORCES AFFECT LABOUR DEMAND

- **Replacement:** machines perform some human tasks.
- **Complementarity:** machines make other human tasks more productive.
- **Expansion:** lower costs can increase sales and exports.

Wages depend on the balance of these forces and on adjustment and institutions.

A Canadian export example

An auto plant can use fewer assembly hours per vehicle and still expand exports. Demand for maintenance or engineering may rise while some assembly tasks contract.

This is a technology extension. Stolper–Samuelson describes goods-price changes with technology fixed.

Source: Task framework: Acemoglu and Restrepo, JEP (2019), Automation and New Tasks. Auto example is a thought experiment.

Canadian AI exposure reaches professional work too

Employment by potential AI exposure and complementarity, March 2026, ages 15–69:

31.2%
high exposure and high complementarity, such as engineers and teachers

29.6%
high exposure and low complementarity, including accounting and software development

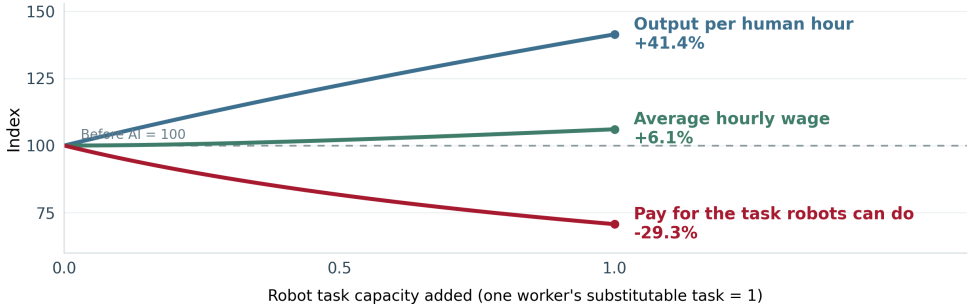
39.3%
low exposure, including many skilled trades and first responders

Adoption is already measurable

35.9% reported using generative AI at work during the previous twelve months. Exposure describes potential task transformation. It does not measure job losses or predict a wage change.

Source: Statistics Canada, July 30, 2026 release, March 2026 LFS supplement, provinces. Categories sum to 100.1% because of rounding.

Productivity can rise while some wages fall



Robots add output. Pay falls for the task they replace and rises for the human task they complement.

Illustration, not Canadian estimates: unchanged human hours and output price. Each input earns its marginal product. Robot services receive part of the extra income.

Source: Course task model. Equations, assumptions and reproducible data.

Canadian robot adopters expand, but workers differ

BANK OF CANADA, SEPTEMBER 2026

Robot adoption in Canadian private corporations is associated with higher revenue, value added, non-owner employment and payroll. The study finds little evidence of a lower labour share and no increase in the total owner income share. The declining-share outcome in the illustration is therefore not a universal prediction.

STATISTICS CANADA, HISTORICAL EVIDENCE

Robot adopters tended to hire more workers overall but needed fewer managers. Middle-skilled workers faced weaker outcomes than high- and low-skilled workers.

Payroll and hourly wages are different

Payroll = total hours × average hourly pay. More payroll does not establish that every worker earns more.

Source: Galassi and MacKenzie, Bank of Canada WP 2026-34; Statistics Canada, Dixon (2020). Firm growth does not by itself identify economy-wide job effects.

YOUR MOVE What worker-level evidence would you need before claiming that automation raised everyone's living standard?

Concentrated losses can dominate political attention

DIFFUSE CONSUMER GAIN

$$10{,}000{,}000 \times \$5 = \$\mathbf{50} \text{ million}$$

Each consumer gains only $5, so few spend time organizing around the policy.

CONCENTRATED PRODUCER LOSS

$$1{,}000 \times \$20{,}000 = \$\mathbf{20} \text{ million}$$

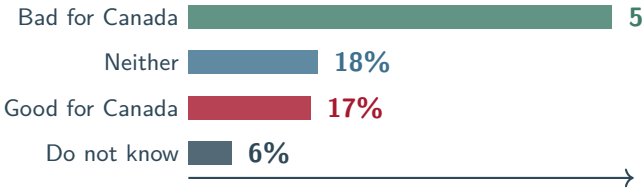
Each affected worker or owner has a large stake and strong reason to organize.

Net measured gain: $\$50\text{m} - \$20\text{m} = \$30\text{m}$

Illustration only: transition costs, taxes, uncertainty, and community spillovers are omitted.

YOUR MOVE Why can the policy face intense opposition even when the aggregate calculation is positive?

Most Canadians viewed growing protectionism as bad in 2021



INTERPRETATION

Canada had a broadly pro-trade baseline. Distributional conflict does not imply broad rejection of trade.

What can change attitudes

Personal exposure, local risk, uncertainty, and confidence in adjustment institutions.

Source: Global Affairs Canada, Canadian Attitudes towards International Trade, February 2021 survey, (n=2,043).

YOUR MOVE Why can a minority position still receive substantial political attention?

Support for protection can be economically intelligible

REASONS FOR CONCERN

- large losses concentrated in one household or town
- specialized skills and assets cannot move quickly
- credit, housing, family, and licensing constrain mobility
- future market access and compensation are uncertain

WHAT DOES NOT FOLLOW

- every tariff raises national income
- protected firms will become competitive
- downstream input costs are negligible
- retaliation and lobbying can be ignored

Understanding the demand for protection is different from concluding that protection is the best remedy.

YOUR MOVE Name one policy that could address worker adjustment without taxing all imports.

The policy instrument should match the underlying problem

Problem	More targeted response	Test before acting
temporary earnings loss	income insurance and rapid re-employment support	does assistance improve stable re-employment?
obsolete or mismatched skills	employer-linked training and credential recognition	are jobs available after training?
immobile local assets	place-based transition and infrastructure with an expiry rule	does private activity persist after support ends?
learning or coordination failure	temporary, conditional industrial support	are productivity and exports rising on schedule?
security dependence	diversify suppliers, inventories, or capacity	how costly is disruption and how quickly can buyers switch?
unfair foreign restriction	narrow, reviewable response and negotiation	who bears domestic input and retaliation costs?

Openness remains the benchmark. A restriction needs a diagnosed market or institutional failure and a measurable exit condition.

YOUR MOVE Which response addresses distribution without trying to reverse comparative advantage?

A current sectoral shock shows where H–O needs evidence

15%

of Canadian exports came from industries facing sectoral U.S. restrictions in early 2026

about 1%

of Canadian output and employment came from those industries

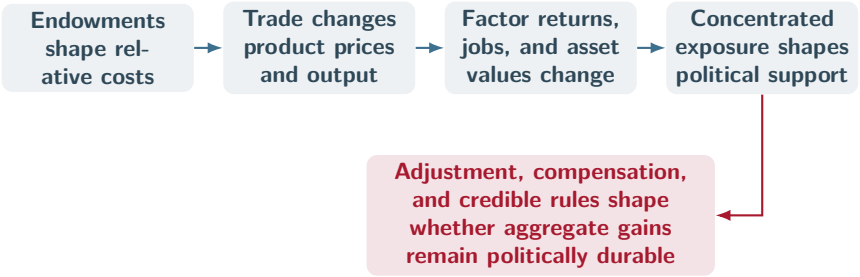
H–O BENCHMARK PLUS ADJUSTMENT EVIDENCE

- a foreign restriction reduces the Canadian producer price or quantity sold; H–O uses factor intensity to predict factor-return pressure
- plants, equipment, supplier networks, and workers can absorb early losses when adjustment is slow
- downstream users and firms can face effects different from primary producers
- distance limits rapid redirection to new markets

Source: Bank of Canada, Monetary Policy Report, April 2026. Output is value added; exports are gross values.

YOUR MOVE Which arrows come from H–O and Stolper–Samuelson? Which require evidence on firms, regions, input links, and gravity?

The complete Week 8 mental model



YOUR MOVE Where would an import tariff enter the chain, and which later boxes determine its political appeal?

Case-reading recipe

When a trade policy changes, ask five questions in order:

1. **Which product price or market access changes?**
2. **Which factors and sectors are used intensively?**
3. **Which ownership patterns and adjustment frictions slow mobility?**
4. **Who gains, who loses, and what happens downstream?**
5. **Can a more targeted institution address the problem at lower social cost?**

Apply it

A foreign tariff reduces demand for Canadian aluminum. Identify the Canadian producer-price effect, the short-run owners and workers exposed, the downstream users, and one adjustment policy.

Week 8 summary

1. H–O links relative factor abundance to the factor intensity of exports.
2. A change in goods prices changes real factor returns; the abundant factor can gain while the scarce factor loses.
3. H–O is a long-run benchmark; observed adjustment can be sectoral, regional, and persistent when mobility is slow.
4. Canadian liberalization combined productivity gains with concentrated adjustment. Long-run mobility helped offset average losses.
5. Concentrated losses help explain support for protection without proving that protection is efficient.
6. Adjustment and industrial policies need a diagnosed problem, evidence, and an exit test.
7. AI can raise productivity while changing wages differently across tasks. Canadian evidence must test who actually gains.

The gains from trade are more likely to remain politically durable when institutions address who bears the adjustment.